

# K8S-99: Best Practice Summary

Series: K8S — Kubernetes Monitoring | Notebook: 99 | Created: March 2026 | Last Updated: 10/02/2026

## Overview

This notebook consolidates every actionable best practice for Dynatrace Kubernetes monitoring and DynaKube configuration extracted from the K8S series (notebooks 01-14). Each practice specifies the exact setting, value, priority, and category. Use this as a definitive checklist for new deployments and audits of existing environments.

**Operator Version:** 1.10.2 (validated pin in this series; 1.11.0, released 10/01/2026, is the newest — see the note below) | **DynaKube API:** `dynatrace.com/v1beta6` — current, use for new DynaKubes ( `v1beta5` remains accepted; no rewrite required) | **Helm:** `oci://public.ecr.aws/dynatrace/dynatrace-operator --version 1.10.2` — skip 1.10.0 (pin the version you have validated in your estate)

**Operator version policy (as of 08/11/2026). 1.10.2 is the recommendation** — released July 30, 2026 with a published changelog. It resolves four issues: a workload/namespace **tagging-precedence regression**, a `dynatrace-webhook` `CrashLoopBackOff` under the gVisor runtime class, injected pods hanging on the OneAgent-binary download (timeout raised to 15 minutes), and metadata-enrichment rules that could not be applied being silently disregarded rather than logged.

**The tagging fix is a behaviour change, not just a bug fix.** Where several tagging rules match the same key, 1.10.2 applies **only the first matching rule**; earlier versions let each subsequent rule overwrite the previous one. A cluster whose enrichment depends on last-rule-wins ordering will produce different tags after the upgrade — audit your rules first. See K8S-10.

**1.10.1 remains a working pin.** Estates upgrade operators on their own schedule, and 1.10.1 is still the fix for the 1.10.0 defects — subject to the OpenShift-manifest caveat in K8S-09 §2. **Skip 1.10.0** — its own release notes advise skipping it (auto-update defect) and it is flagged `prerelease: true` on the [GitHub releases page](#), a signal you can verify before pinning. Do not pin below 1.4.1 in any case (CSI liveness-probe crash-loop window — K8S-09 §2).

**Operator 1.11.0 (released 10/01/2026).** Removes the `v1beta4` DynaKube API from the CRD (“Applying DynaKube resources that still use `v1beta4` will fail”), adds image-volume code-module injection, which “replaces the CSI driver as the recommended approach” (Kubernetes 1.35+ and containerd 2.2+ or CRI-O 1.33+ required; rule 7), adds `spec.kubernetesMonitoring` to size the Kubernetes-monitoring ActiveGate separately in one DynaKube, and accepts workload and pod labels and annotations as enrichment sources (rules 39–41). Minimum version for a direct upgrade: 1.6.0. Move the pin once you have validated 1.11.0 on a non-production cluster.

**Token currency note:** Platform Tokens ( `dt0s16` ) are the recommended choice for new tenants per Dynatrace SaaS sprint-1.337+; the Operator itself accepts platform tokens from **Operator 1.10.0** (July 15, 2026) — Classic API Tokens ( `dt0c01` ) continue to be accepted and remain the working path on earlier Operator versions. See K8S-02 §Prerequisites.

**Operator 1.10.0 upgrade notes (July 15, 2026):** the ActiveGate `TopologySpreadConstraint` default changed from `DoNotSchedule` to `ScheduleAnyway` — expect an **ActiveGate restart on upgrade** (an explicitly-set `whenUnsatisfiable` wins, so the recommended posture is to set it; see K8S-02 §8). DynaKube `v1beta3` was removed in Operator 1.9.0, and **`v1beta4` stopped being served in 1.10.0** (it was deprecated in 1.9.0) — so a `v1beta4` manifest fails to apply after this upgrade, not merely warns. `v1beta5` is still served but is itself flagged deprecated from 1.10.0. Target `v1beta6` for new DynaKubes. The `gke-autopilot.yaml` release artifact is removed — GKE Autopilot deploys via Helm. Two init containers were introduced — `webhook-cert-generator` and `crd-storage-migrator` — which is what an `Init:0/1` hang is waiting on (K8S-09 §2). 1.10.0 also adds a **CSI-to-ephemeral-volume migration mode**, which turns code-module delivery into a deliberate choice rather than a fixed default — see rule 7 and K8S-12 §2.

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## Prerequisites

| Requirement           | Details                                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dynatrace Environment | SaaS with Grail and Kubernetes monitoring enabled                                                                                                                                                              |
| Kubernetes Cluster    | A version supported by your Operator release                                                                                                                                                                   |
| Helm                  | v3.x                                                                                                                                                                                                           |
| Dynatrace Operator    | v1.10.2 (July 30, 2026, recommended pin) via <code>oci://public.ecr.aws/dynatrace/dynatrace-operator</code> — v1.10.1 remains a working pin until you upgrade; skip 1.10.0 (see the version-policy note above) |
| Knowledge             | K8S-01 through K8S-13                                                                                                                                                                                          |

## 1. Deployment Mode Selection

| # | Best Practice                                                                                    | Recommended Setting/Value                                                                                                | Priority    | Category   |
|---|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 1 | Use <code>cloudNativeFullStack</code> for new deployments                                        | <code>spec.oneAgent.cloudNativeFullStack: {}</code>                                                                      | Critical    | Deployment |
| 2 | Do NOT use <code>classicFullStack</code> for new deployments                                     | Remove <code>spec.oneAgent.classicFullStack</code>                                                                       | Critical    | Deployment |
| 3 | Use <code>applicationMonitoring</code> when infra visibility is not needed                       | <code>spec.oneAgent.applicationMonitoring: {}</code> (CSI driver use is chosen at Operator install, not in the DynaKube) | Recommended | Deployment |
| 4 | Use <code>hostMonitoring</code> for infra-only clusters                                          | <code>spec.oneAgent.hostMonitoring: {}</code>                                                                            | Optional    | Deployment |
| 5 | Omit <code>oneAgent</code> entirely for infrastructure-only monitoring alongside other APM tools | Only configure <code>spec.activeGate</code>                                                                              | Recommended | Deployment |

**classicFullStack** is legacy. It uses host-path mounts and shares a single OneAgent across all pods. `cloudNativeFullStack` injects code modules via the webhook — delivered by image volume, CSI driver or ephemeral volume (rule 7) — enabling

independent app/infra monitoring with no privileged containers for applications.

## 2. Operator Installation

| #  | Best Practice                                                                 | Recommended Setting/Value                                                                                                                                                                                                                                                                                                    | Priority                              | Category     |
|----|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------|
| 6  | Install operator via Helm OCI, always with an explicit <code>--version</code> | <pre>helm upgrade dynatrace-operator oci://public.ecr.aws/dynatrace/dynatrace-operator --version 1.10.2 --namespace dynatrace --create-namespace --install --atomic</pre>                                                                                                                                                    | Critical                              | Installation |
| 7  | Choose a code-module delivery mode deliberately                               | Image volumes on Operator 1.11.0+ where nodes meet the requirements (Dynatrace's recommended approach); otherwise <code>csidriver.enabled: true</code> ; ephemeral volumes when a privileged DaemonSet is ruled out (Operator 1.10.0+). To leave CSI, migrate first — never flip <code>csidriver.enabled: false</code> first | Critical ( <i>making the choice</i> ) | Installation |
| 8  | Set platform explicitly                                                       | <code>platform: "kubernetes"</code> or <code>platform: "openshift"</code>                                                                                                                                                                                                                                                    | Recommended                           | Installation |
| 9  | Create dedicated namespace                                                    | <pre>kubectl create namespace dynatrace</pre>                                                                                                                                                                                                                                                                                | Critical                              | Installation |
| 10 | Create API token secret before applying DynaKube                              | <pre>kubectl create secret generic dynakube --namespace dynatrace -- from-literal=apiToken=&lt;TOKEN&gt; --from-literal=dataIngestToken= &lt;TOKEN&gt;</pre>                                                                                                                                                                 | Critical                              | Installation |

**Rule 7 — code-module delivery is a choice, not an invariant.** Through Operator 1.9.x, `csidriver.enabled: true` was effectively the only production answer. **Operator 1.10.0 adds a CSI-to-ephemeral-volume migration mode**, and **Operator 1.11.0 adds image volumes**, which Dynatrace now recommends over the CSI driver. What is Critical is *making the decision consciously* — not one specific value.

| Delivery mode                                                                                                                          | Choose it when                                                                                                                                                           | Cost                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Image volumes</b> (Operator 1.11.0+; DynaKube flag <code>feature.dynatrace.com/mount-code-modules-via-image-volume: "true"</code> ) | Dynatrace's recommended approach where the cluster runs Kubernetes 1.35+ with containerd 2.2+ or CRI-O 1.33+. Not compatible with the built-in tenant registry.          | Existing pods keep their mounts until restarted, so a full switch needs a rolling restart.                                                 |
| <b>CSI driver</b> ( <code>csidriver.enabled: true</code> )                                                                             | The image-volume requirements are not met. You want code modules cached per node and shared across pods, and you can run a privileged DaemonSet with a host-path socket. | An extra 5-container DaemonSet to size and monitor; a mount-storm failure mode (rules 57–59, K8S-09 §2).                                   |
| <b>Ephemeral volumes</b> (Operator 1.10.0+)                                                                                            | A CSI DaemonSet is unacceptable — restrictive admission policy, a managed platform that limits CSI drivers, or a node pool where you will not run privileged workloads.  | Code modules are provisioned per pod rather than shared per node, so expect more image/volume churn and slower pod starts at high density. |

Migrating CSI → ephemeral (the documented direction) is a Helm-values change plus one workload-restart cycle, not an application change: enable `csidriver.migrationMode`, restart injected workloads, confirm no pod still mounts `csi.oneagent.dynatrace.com`, and only then set `csidriver.enabled: false` — pods still on CSI mounts stop working once the driver is disabled. Operator 1.10.0 was released July 15, 2026 and estates adopt it on their own schedule — **on Operator 1.9.x and earlier the CSI driver remains the only supported mode**, so `csidriver.enabled: true` stays the correct setting there. Mechanics and trade-offs: K8S-12 §2–§3.

## Required Token Scopes

The docs describe platform tokens (Operator 1.10.0+); classic access tokens remain accepted. Full tables: K8S-02 § Prerequisites.

**Operator token:** `fleet-management:activegate.connection-info:read`, `fleet-management:activegate.tokens:create`, `fleet-management:container-images:read`, `fleet-management:oneagent.connection-info:read`, `fleet-management:oneagents:download`, `settings:objects:read`, `settings:objects:write`

**Data Ingest token:** `openpipeline:logs:ingest`, `openpipeline:metrics:ingest`, `openpipeline:traces:ingest`, `storage:metrics:write`

## 3. DynaKube Core Configuration

| #  | Best Practice                                      | Recommended Setting/Value                                                                                                                                                                                                                                                              | Priority    | Category      |
|----|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| 11 | Use <code>v1beta6</code> for new DynaKubes         | <code>apiVersion: dynatrace.com/v1beta6</code> . <code>v1beta5</code> still applies but is <b>deprecated from 1.10.0</b> ; <code>v1beta4</code> is <b>not served from 1.10.0</b> and <b>removed from the CRD in 1.11.0</b> (apply fails); <code>v1beta3</code> <b>removed</b> in 1.9.0 | Critical    | Configuration |
| 12 | Set <code>apiUrl</code> to your SaaS tenant        | <code>spec.apiUrl: https://ENVIRONMENT_ID.live.dynatrace.com/api</code>                                                                                                                                                                                                                | Critical    | Configuration |
| 13 | Add control-plane tolerations to OneAgent          | <code>tolerations: [{effect: NoSchedule, key: node-role.kubernetes.io/master, operator: Exists}, {effect: NoSchedule, key: node-role.kubernetes.io/control-plane, operator: Exists}]</code>                                                                                            | Recommended | Configuration |
| 14 | Set <code>nodeSelector</code> for linux-only       | <code>nodeSelector: {kubernetes.io/os: linux}</code>                                                                                                                                                                                                                                   | Recommended | Configuration |
| 15 | Set OneAgent resource requests and limits          | <code>spec.oneAgent.cloudNativeFullStack.oneAgentResources</code> — e.g. <code>requests: {cpu: 100m, memory: 256Mi}</code> , <code>limits: {cpu: 300m, memory: 512Mi}</code> as a community starting point; size from measured usage                                                   | Recommended | Configuration |
| 16 | Set <code>networkZone</code> for routing isolation | <code>spec.networkZone: production</code>                                                                                                                                                                                                                                              | Optional    | Configuration |
| 17 | Set <code>hostGroup</code> for logical grouping    | <code>spec.oneAgent.hostGroup: production</code>                                                                                                                                                                                                                                       | Optional    | Configuration |

Minimal Production DynaKube

```
apiVersion: dynatrace.com/v1beta6
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://ENVIRONMENT_ID.live.dynatrace.com/api
  metadataEnrichment:
    enabled: true
  oneAgent:
    cloudNativeFullStack:
      tolerations:
        - effect: NoSchedule
          key: node-role.kubernetes.io/master
          operator: Exists
        - effect: NoSchedule
          key: node-role.kubernetes.io/control-plane
          operator: Exists
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
      - dynatrace-api
```

4. ActiveGate Configuration

| #  | Best Practice                                        | Recommended Setting/Value                                                                                                           | Priority    | Category   |
|----|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 18 | Enable <code>kubernetes-monitoring</code> capability | <code>spec.activeGate.capabilities: [kubernetes-monitoring, routing]</code>                                                         | Critical    | ActiveGate |
| 19 | Add <code>dynatrace-api</code> capability            | Add <code>dynatrace-api</code> to capabilities list                                                                                 | Recommended | ActiveGate |
| 20 | Set ActiveGate replicas $\geq 2$ for production      | <code>spec.activeGate.replicas: 2</code>                                                                                            | Critical    | ActiveGate |
| 21 | Set ActiveGate resource limits                       | e.g. <code>limits: {cpu: 1000m, memory: 2Gi}</code> for medium clusters (community starting point; FAQ-10 for sizing)               | Critical    | ActiveGate |
| 22 | Add zone-aware topology spread                       | <code>topologySpreadConstraints: [{maxSkew: 1, topologyKey: topology.kubernetes.io/zone, whenUnsatisfiable: ScheduleAnyway}]</code> | Recommended | ActiveGate |

ActiveGate Sizing Reference

Starting points from community practice, not Dynatrace-published figures — measure and adjust (FAQ-10).

| Cluster Size | Nodes  | CPU Limit | Memory Limit | Replicas |
|--------------|--------|-----------|--------------|----------|
| Small        | 1-10   | 500m      | 1Gi          | 1        |
| Medium       | 10-50  | 1000m     | 2Gi          | 2        |
| Large        | 50-100 | 2000m     | 4Gi          | 2        |
| Enterprise   | 100+   | 4000m     | 8Gi          | 3+       |

## 5. Namespace and Injection Control

| #  | Best Practice                                                  | Recommended Setting/Value                                                                                                                                    | Priority    | Category  |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 23 | Use <code>namespaceSelector</code> to control injection scope  | <code>spec.oneAgent.cloudNativeFullStack.namespaceSelector.matchLabels: {dynatrace-injection: enabled}</code>                                                | Recommended | Injection |
| 24 | Do not use the default namespace for workloads                 | Enforce via policy                                                                                                                                           | Recommended | Namespace |
| 25 | Apply consistent labels to all namespaces                      | <code>team</code> , <code>env</code> , <code>cost-center</code> labels                                                                                       | Recommended | Namespace |
| 26 | Use <code>matchExpressions</code> to exclude system namespaces | <code>operator: NotIn</code> , <code>values: [kube-system, newrelic, datadog]</code>                                                                         | Recommended | Injection |
| 27 | Use pod annotation to disable injection for specific pods      | <code>oneagent.dynatrace.com/inject: "false"</code>                                                                                                          | Optional    | Injection |
| 28 | Limit injected technologies when needed                        | <code>oneagent.dynatrace.com/technologies: "java,nodejs"</code> — <i>"Ignored if the CSI volume is used or node image pull via ephemeral volume is used"</i> | Optional    | Injection |
| 29 | Set ResourceQuotas on every namespace                          | <code>requests.cpu</code> , <code>requests.memory</code> , <code>limits.cpu</code> , <code>limits.memory</code> , <code>pods</code>                          | Recommended | Namespace |
| 30 | Set LimitRanges for default container limits                   | <code>default: {cpu: 500m, memory: 512Mi}</code> , <code>defaultRequest: {cpu: 100m, memory: 128Mi}</code>                                                   | Recommended | Namespace |

## 6. Feature Flags and Annotations

| #  | Best Practice                                                                                            | Recommended Setting/Value                                                                                                                                                                                                                                    | Priority    | Category     |
|----|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|
| 31 | Do not set the obsolete <code>k8s-app-enabled</code> flag                                                | The docs: <i>"The schema is no longer available on newer Dynatrace environments, where the Kubernetes app experience is enabled automatically."</i>                                                                                                          | Recommended | Feature Flag |
| 32 | Enable version detection from K8s labels                                                                 | <code>feature.dynatrace.com/label-version-detection: "true"</code>                                                                                                                                                                                           | Recommended | Feature Flag |
| 33 | Set injection failure policy to <code>fail</code> in non-prod                                            | <code>feature.dynatrace.com/injection-failure-policy: "fail"</code>                                                                                                                                                                                          | Recommended | Feature Flag |
| 34 | Set injection failure policy to <code>silent</code> in prod                                              | <code>feature.dynatrace.com/injection-failure-policy: "silent"</code>                                                                                                                                                                                        | Critical    | Feature Flag |
| 35 | Use opt-in for multi-tool coexistence                                                                    | Namespace opt-in: <code>namespaceSelector</code> alone. Pod opt-in: <code>feature.dynatrace.com/automatic-injection: "false"</code> <b>plus</b> <code>oneagent.dynatrace.com/inject: "true"</code> on each pod — the flag alone injects nothing (K8S-11 § 2) | Recommended | Feature Flag |
| 36 | Apply <code>app.kubernetes.io/version</code> and <code>app.kubernetes.io/part-of</code> to pod templates | Read as <code>DT_RELEASE_VERSION / DT_RELEASE_PRODUCT</code> ; only pod labels count                                                                                                                                                                         | Recommended | Labels       |

| #  | Best Practice                                         | Recommended Setting/Value                                            | Priority | Category |
|----|-------------------------------------------------------|----------------------------------------------------------------------|----------|----------|
| 37 | Set the release stage label where you want it tracked | <code>dynatrace-release-stage</code> → <code>DT_RELEASE_STAGE</code> | Optional | Labels   |

## 7. Metadata Enrichment

| #  | Best Practice                                                                                      | Recommended Setting/Value                                                                                                                                                                                                 | Priority    | Category   |
|----|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 38 | Enable metadata enrichment where OneAgent does not inject                                          | <code>spec.metadataEnrichment.enabled: true</code> ( <code>false</code> by default in <code>v1beta6</code> ; OneAgent injection enriches injected pods on its own)                                                        | Critical    | Enrichment |
| 39 | Choose the enrichment method by where the value is maintained                                      | Central rules for existing namespace labels; <code>metadata.dynatrace.com/&lt;key&gt;</code> annotations for per-workload or per-pod values; DynaKube <code>resourceAttributes</code> for cluster-wide facts (K8S-10 § 2) | Critical    | Enrichment |
| 40 | Know the precedence when two methods set the same key                                              | Annotations > DynaKube attributes > central rules; within annotations, pod > workload > namespace                                                                                                                         | Critical    | Enrichment |
| 41 | Create enrichment rules for <code>team</code> , <code>env</code> , <code>cost-center</code> labels | Metadata type <code>Label</code> , Source <code>cost-center</code> , Target <code>dt.cost.costcenter</code> (or "enrich directly" for a <code>k8s.namespace.label.&lt;key&gt;</code> field)                               | Recommended | Enrichment |
| 42 | Check version floors before relying on a level                                                     | Kubernetes enrichment: Operator 1.10+, OneAgent 1.333+, ActiveGate 1.343+; workload-level annotations: Operator 1.11.0+, ActiveGate 1.349+                                                                                | Recommended | Enrichment |
| 43 | Wait, then restart, after rule changes                                                             | <i>"New rules may take up to 45 minutes to take effect. Pod restarts are required after the 45 mins"</i>                                                                                                                  | Recommended | Enrichment |
| 44 | Label namespaces with cost allocation metadata                                                     | <code>cost-center</code> , <code>budget-owner</code> labels on Namespace objects                                                                                                                                          | Recommended | Enrichment |

## 8. Log Monitoring

| #  | Best Practice                                                   | Recommended Setting/Value                                                                                                                   | Priority    | Category |
|----|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| 45 | Enable log monitoring                                           | <code>spec.logMonitoring: {}</code> , optionally with <code>ingestRuleMatchers</code> (immutable once set)                                  | Recommended | Logs     |
| 46 | Configure log monitoring resources under <code>templates</code> | <code>spec.templates.logMonitoring.resources: {limits: {cpu: 500m, memory: 512Mi}}</code>                                                   | Critical    | Logs     |
| 47 | Keep pod settings out of <code>spec.logMonitoring</code>        | Its only field is <code>ingestRuleMatchers</code> ; resources and tolerations go under <code>spec.templates.logMonitoring</code>            | Critical    | Logs     |
| 48 | Add control-plane tolerations to log monitoring                 | <code>spec.templates.logMonitoring.tolerations: [{effect: NoSchedule, key: node-role.kubernetes.io/control-plane, operator: Exists}]</code> | Recommended | Logs     |
| 49 | Use OpenPipeline to filter debug logs before storage            | Drop <code>loglevel == "DEBUG"</code> in processing pipeline                                                                                | Recommended | Logs     |
| 50 | Store logs in separate Grail buckets by environment             | OpenPipeline Bucket assignment processors matching an enriched field such as <code>k8s.namespace.label.env</code>                           | Optional    | Logs     |

## 9. OTEL Collector and Telemetry Ingest

| #  | Best Practice                                                                                                                                                        | Recommended Setting/Value                                                                                                                                                      | Priority    | Category |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| 51 | Pin OTEL Collector image to a specific, verified version (example: <code>0.57.0</code> , released 09/24/2026 — check the <a href="#">releases</a> before installing) | <code>spec.templates.otelCollector.imageRef.tag: "0.57.0"</code>                                                                                                               | Critical    | OTel     |
| 52 | Never use <code>latest</code> tag for OTel Collector                                                                                                                 | Always specify explicit version tag                                                                                                                                            | Critical    | OTel     |
| 53 | Set OTel Collector resource limits                                                                                                                                   | <code>limits: {cpu: 500m, memory: 512Mi}</code> for staging; <code>{cpu: 1000m, memory: 1Gi}</code> for production                                                             | Recommended | OTel     |
| 54 | Configure <code>telemetryIngest</code> with required protocols                                                                                                       | <code>spec.telemetryIngest.protocols: [otlp]</code> (add <code>statsd</code> , <code>jaeger</code> , <code>zipkin</code> as needed)                                            | Recommended | OTel     |
| 55 | Do not use the OneAgent StatsD daemon on Kubernetes                                                                                                                  | Use an environment ActiveGate as remote listener (the docs' recommendation), DynaKube <code>telemetryIngest</code> with <code>statsd</code> , or a self-managed OTel Collector | Critical    | OTel     |
| 56 | If you run your own StatsD collector, give it a dedicated namespace                                                                                                  | Cluster-wide Deployment; apps reference via FQDN <code>otel-collector-statsd.monitoring.svc.cluster.local:8125</code>                                                          | Recommended | OTel     |

## 10. CSI Driver Configuration

| #  | Best Practice                                                              | Recommended Setting/Value                                                                                                   | Priority                            | Category |
|----|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------|
| 57 | Decide image volumes vs. CSI vs. ephemeral volumes, then set it explicitly | See rule 7 — image volumes are recommended from Operator 1.11.0 where nodes qualify                                         | Critical <i>(making the choice)</i> | CSI      |
| 58 | Set resource limits on provisioner container                               | <code>provisioner.resources.limits: {cpu: 500m, memory: 256Mi}</code>                                                       | Critical                            | CSI      |
| 59 | Set resource limits on all 5 CSI containers                                | <code>csiInit</code> , <code>server</code> , <code>provisioner</code> , <code>registrar</code> , <code>livenessprobe</code> | Recommended                         | CSI      |

### CSI Driver Container Resource Reference

| Container                  | Chart default (Operator 1.11.0) requests / limits | Suggested (community starting point) requests / limits |
|----------------------------|---------------------------------------------------|--------------------------------------------------------|
| <code>csiInit</code>       | 50m, 100Mi / 50m, 100Mi                           | 50m, 100Mi / 100m, 128Mi                               |
| <code>server</code>        | 50m, 100Mi / 50m, 100Mi                           | 50m, 100Mi / 100m, 128Mi                               |
| <code>provisioner</code>   | 300m, 100Mi / <b>none</b>                         | 300m, 100Mi / 500m, 256Mi                              |
| <code>registrar</code>     | 20m, 30Mi / 20m, 30Mi                             | 20m, 30Mi / 50m, 64Mi                                  |
| <code>livenessprobe</code> | 20m, 30Mi / 20m, 30Mi                             | 20m, 30Mi / 50m, 64Mi                                  |

**Warning:** In the Operator 1.11.0 Helm chart the `provisioner` container has requests but no limits. Add them explicitly if your policies require limits.

**Rules 58–59 apply only on the CSI path.** If you chose image volumes (Operator 1.11.0+) or ephemeral volumes (Operator 1.10.0+, see rule 7) there is no CSI DaemonSet to size, and this entire section is not applicable — the corresponding work moves to per-pod volume provisioning and pod-start latency at high density. On Operator 1.9.x and earlier the CSI path is the only supported mode, so these rules always apply.

## 11. Resource Sizing

| #  | Best Practice                                                       | Recommended Setting/Value                                                                                                                                                                                       | Priority    | Category |
|----|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| 60 | Set OneAgent resources for cloudNativeFullStack                     | <code>oneAgentResources</code> — e.g. <code>requests: {cpu: 100m, memory: 256Mi}</code> , <code>limits: {cpu: 300m, memory: 512Mi}</code> (community starting point)                                            | Recommended | Sizing   |
| 61 | Size ActiveGate by cluster node count                               | See ActiveGate Sizing Reference in Section 4                                                                                                                                                                    | Critical    | Sizing   |
| 62 | Size OTel Collector by telemetry volume (community starting points) | Low: 200m/256Mi, Medium: 500m/512Mi, High: 1000m/1Gi, Very High: 2000m/2Gi                                                                                                                                      | Recommended | Sizing   |
| 63 | Size Log Monitoring by daily log volume (community starting points) | Low (<1GB): 200m/256Mi, Medium (1-10GB): 500m/512Mi, High (10-50GB): 1000m/1Gi                                                                                                                                  | Recommended | Sizing   |
| 64 | Monitor OneAgent memory by absolute usage                           | OneAgent pods have limits only if you set <code>oneAgentResources</code> ; use <code>dt.kubernetes.container.memory_working_set</code> with <code>matchesValue(k8s.container.name, "dynatrace-oneagent")</code> | Recommended | Sizing   |

## 12. Security and Secrets

| #  | Best Practice                                               | Recommended Setting/Value                                                                                                                                  | Priority    | Category |
|----|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| 65 | Never store API tokens in Git                               | Use External Secrets Operator, Sealed Secrets, or SOPS                                                                                                     | Critical    | Security |
| 66 | Use External Secrets Operator for token management          | <code>ExternalSecret</code> CR referencing Vault, AWS Secrets Manager, or GCP Secret Manager                                                               | Recommended | Security |
| 67 | Apply NetworkPolicies allowing Dynatrace egress on port 443 | <code>egress: [{to: [{ipBlock: {cidr: 0.0.0.0/0}}], ports: [{protocol: TCP, port: 443}]}]</code>                                                           | Recommended | Security |
| 68 | Configure proxy via DynaKube spec (not env vars)            | <code>spec.proxy.value: https://proxy.example.com:8080</code> or <code>spec.proxy.valueFrom.secretKeyRef</code>                                            | Recommended | Security |
| 69 | Mirror Kubernetes namespace RBAC in Dynatrace IAM           | A namespace bound to <code>view / edit / admin</code> maps to a read policy with <code>WHERE storage:k8s.namespace.name = "&lt;ns&gt;"</code> (K8S-06 § 4) | Recommended | Security |

## 13. GitOps and Lifecycle

| #  | Best Practice                                                  | Recommended Setting/Value                                                                                                                                                                                                                                                               | Priority    | Category  |
|----|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 70 | Manage DynaKube via GitOps (ArgoCD or Flux)                    | Store DynaKube YAML in Git, apply via GitOps controller                                                                                                                                                                                                                                 | Recommended | Lifecycle |
| 71 | Use Kustomize overlays for environment-specific config         | <code>base/dynakube.yaml</code> + <code>overlays/{dev,staging,prod}/</code> patches                                                                                                                                                                                                     | Recommended | Lifecycle |
| 72 | Pin an <b>exact</b> operator chart version in ArgoCD/Flux      | <code>targetRevision: 1.10.2</code> (ArgoCD) or <code>version: "1.10.2"</code> (Flux) — prefer an exact pin over a <code>1.10.x</code> range, because an operator upgrade can change chart <i>defaults</i> with no Git commit to review (K8S-02 §8). Never pin below 1.4.1; skip 1.10.0 | Critical    | Lifecycle |
| 73 | Set <code>selfHeal: true</code> in ArgoCD for drift correction | <code>syncPolicy.automated.selfHeal: true</code>                                                                                                                                                                                                                                        | Recommended | Lifecycle |
| 74 | Set <code>prune: false</code> for DynaKube in ArgoCD           | Prevent accidental DynaKube deletion on Git removal                                                                                                                                                                                                                                     | Critical    | Lifecycle |

| #  | Best Practice                                                                                           | Recommended Setting/Value                                                                                                                                                                                                             | Priority    | Category  |
|----|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 75 | Use Flux <code>dependsOn</code> to order operator before DynaKube                                       | <code>dependsOn: [{name: dynatrace-operator}]</code>                                                                                                                                                                                  | Recommended | Lifecycle |
| 76 | Use ArgoCD sync waves for deployment ordering                                                           | Wave 0: Namespace, Wave 1: Operator, Wave 2: DynaKube                                                                                                                                                                                 | Recommended | Lifecycle |
| 77 | Use <code>helm upgrade --reuse-values</code> for operator upgrades — <b>and diff the rendered chart</b> | Preserves <i>your</i> values, but <b>not</b> changed chart <i>defaults</i> for values you never set. Render and diff before upgrading ( <code>helm template ... &gt; new.yaml</code> vs. <code>helm get manifest</code> ) — K8S-02 §8 | Recommended | Lifecycle |
| 78 | Deploy to dev -> staging -> prod-canary -> prod                                                         | Progressive rollout strategy                                                                                                                                                                                                          | Recommended | Lifecycle |

## 14. Cluster Health Alerting

| #  | Best Practice                                    | Recommended Setting/Value                                                                                                               | Priority        | Category |
|----|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|
| 79 | Alert on Node NotReady                           | Built-in <i>Detect node readiness issues</i> , or <code>dt.kubernetes.node.conditions Ready = false for &gt;5 min = Critical</code>     | <b>Critical</b> | Alerting |
| 80 | Alert on high node CPU                           | CPU > 85% for 15 min = Warning                                                                                                          | Recommended     | Alerting |
| 81 | Alert on high node memory                        | Memory > 90% for 10 min = Critical                                                                                                      | <b>Critical</b> | Alerting |
| 82 | Alert on disk pressure                           | DiskPressure condition true, or disk > 85% = Warning                                                                                    | Recommended     | Alerting |
| 83 | Alert on OOM kills                               | <code>dt.kubernetes.container.oom_kills &gt; 0</code> = Warning (a metric, not an event)                                                | <b>Critical</b> | Alerting |
| 84 | Alert on restart loops                           | <code>BackOff events</code> or <code>dt.kubernetes.container.restarts</code> = Warning (CrashLoopBackOff is message text, not a reason) | Recommended     | Alerting |
| 85 | Alert on FailedScheduling                        | Events > 5 in 10 min = Warning                                                                                                          | Recommended     | Alerting |
| 86 | Alert on volume mount failures                   | Any FailedMount or FailedAttachVolume = Critical                                                                                        | <b>Critical</b> | Alerting |
| 87 | Monitor Dynatrace component health (OA, AG, CSI) | Track restarts, OOMKills, and data gaps for <code>dynatrace-oneagent</code> and <code>activegate</code> containers                      | <b>Critical</b> | Alerting |
| 88 | Right-size CPU requests                          | Workload CPU used / requested; ~40% average is a community starting point (K8S-04 § 6)                                                  | Recommended     | Cost     |
| 89 | Right-size memory requests                       | Workload working set / requested; ~50% average is a community starting point, keep peak headroom                                        | Recommended     | Cost     |

## 15. Workload and Application Monitoring

| #  | Best Practice                                | Recommended Setting/Value                                                                   | Priority        | Category |
|----|----------------------------------------------|---------------------------------------------------------------------------------------------|-----------------|----------|
| 90 | Alert on container CPU usage > 85% sustained | Monitor <code>dt.kubernetes.container.cpu_usage</code> vs limits                            | Recommended     | Workload |
| 91 | Alert on container memory usage > 90%        | Monitor <code>dt.kubernetes.container.memory_working_set</code> vs limits                   | <b>Critical</b> | Workload |
| 92 | Monitor CPU throttling                       | <code>dt.containers.cpu.throttled_time &gt; 0</code> indicates under-provisioned CPU limits | Recommended     | Workload |
| 93 | Track service P99 latency and error rate     | SLO: P99 < 500ms, error rate < 0.1%                                                         | Recommended     | Workload |

| #  | Best Practice                                                              | Recommended Setting/Value                                       | Priority    | Category |
|----|----------------------------------------------------------------------------|-----------------------------------------------------------------|-------------|----------|
| 94 | Filter on <code>isNotNull(field)</code> before aggregating optional fields | Prevent null group-by keys in DQL                               | Recommended | Workload |
| 95 | Use <code>arrayAvg()</code> before filtering/sorting timeseries results    | <code>timeseries</code> returns arrays; convert to scalar first | Critical    | DQL      |

## 16. Specialized Monitoring

| #   | Best Practice                                                                            | Recommended Setting/Value                                                                                                                                                                                                                   | Priority    | Category   |
|-----|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 96  | Instrument NGINX Ingress via ConfigMap <code>main-snippet</code>                         | <code>load_module /opt/dynatrace/oneagent-paas/agent/bin/current/linux-musl-x86-64/liboneagentnginx.so;</code>                                                                                                                              | Recommended | NGINX      |
| 97  | NGINX instrumentation requires OneAgent 1.227+ and no ARM64                              | Pod or container name must contain <code>ingress-nginx-</code> or <code>nginx-ingress-</code>                                                                                                                                               | Recommended | NGINX      |
| 98  | Enable Prometheus scraping for custom metrics                                            | Annotate pods with <code>metrics.dynatrace.com/scrape: "true"</code> , <code>/port</code> , <code>/path</code>                                                                                                                              | Recommended | Prometheus |
| 99  | Scrape Prometheus from inside the cluster — and prefer the OTel Collector for new setups | An external ActiveGate cannot scrape endpoints that require authentication; the ActiveGate integration has hard limits (1,000 exporter pods, 1,000 metrics per pod), and the docs recommend the OpenTelemetry Collector for new deployments | Critical    | Prometheus |
| 100 | Use Kpow + Prometheus scraping for Kafka consumer lag monitoring                         | <code>PROMETHEUS_EGRESS: "true"</code> in Kpow config; scrape annotations on Kpow pods                                                                                                                                                      | Optional    | Kafka      |

## 17. Troubleshooting Practices

| #   | Best Practice                                   | Recommended Setting/Value                                                                                                                                          | Priority    | Category        |
|-----|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|
| 101 | Check DynaKube status first                     | <code>kubectl -n dynatrace get dynakube</code> — expect <code>Running</code> phase                                                                                 | Critical    | Troubleshooting |
| 102 | Verify all Dynatrace pods are running           | <code>kubectl -n dynatrace get pods</code> — no <code>CrashLoopBackOff</code> or <code>Error</code>                                                                | Critical    | Troubleshooting |
| 103 | Check OneAgent connectivity to tenant           | <code>kubectl -n dynatrace exec &lt;oneagent-pod&gt; -c oneagent -- curl -v https://&lt;tenant&gt;.live.dynatrace.com/api/v1/time</code>                           | Recommended | Troubleshooting |
| 104 | Check namespace labels when injection fails     | <code>kubectl get namespace &lt;ns&gt; -o jsonpath='{.metadata.labels}'</code> must match DynaKube <code>namespaceSelector</code>                                  | Recommended | Troubleshooting |
| 105 | Verify init container present for injected pods | <code>kubectl get pod &lt;pod&gt; -o jsonpath='{.spec.initContainers[*].name}'</code>                                                                              | Recommended | Troubleshooting |
| 106 | Check webhook registration                      | <code>kubectl get mutatingwebhookconfigurations</code> — Dynatrace webhook must exist                                                                              | Recommended | Troubleshooting |
| 107 | Collect a support archive for escalation        | <code>kubectl exec -n dynatrace deployment/dynatrace-operator -- dynatrace-operator support-archive --stdout &gt; operator-support-archive.zip</code> (K8S-09 § 8) | Optional    | Troubleshooting |

| #   | Best Practice                                               | Recommended Setting/Value                                                                                                                                                                     | Priority    | Category        |
|-----|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|
| 108 | Use DQL to detect Dynatrace component failures across fleet | <code>events with event.provider == "KUBERNETES_EVENT" in the dynatrace namespace ( Failed , Backoff , FailedMount ), plus the oom_kills and restarts metrics</code>                          | Recommended | Troubleshooting |
| 109 | Detect metric data gaps from null buckets                   | <code>timeseries ... = avg(dt.host.cpu.usage) then compare arraySize(arrayRemoveNulls(...)) to arraySize(...) — arraySize</code><br>counts empty buckets and <code>arrayMin</code> skips them | Recommended | Troubleshooting |

## Summary

This notebook contains **109 actionable best practices** across 17 categories for Dynatrace Kubernetes monitoring:

| Category               | Count | Key Takeaway                                                                                                     |
|------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| Deployment Mode        | 5     | Use <code>cloudNativeFullStack</code> ; never <code>classicFullStack</code> for new deployments                  |
| Operator Installation  | 5     | Helm OCI install with an explicit version; choose code-module delivery deliberately                              |
| DynaKube Core          | 7     | <code>v1beta6</code> , tolerations, resource limits                                                              |
| ActiveGate             | 5     | <code>kubernetes-monitoring + routing</code> , 2+ replicas in prod                                               |
| Namespace/Injection    | 8     | <code>namespaceSelector</code> for scope control, ResourceQuotas on every namespace                              |
| Feature Flags          | 7     | Version detection, opt-in levels, injection failure policy                                                       |
| Metadata Enrichment    | 7     | Three methods, one precedence order, version floors                                                              |
| Log Monitoring         | 6     | <code>logMonitoring</code> (optionally <code>ingestRuleMatchers</code> ), resources under <code>templates</code> |
| OTel/Telemetry Ingest  | 6     | Pin version; StatsD via environment ActiveGate, telemetry ingest or a collector                                  |
| CSI Driver             | 3     | Provisioner has no default limits; set them on the CSI path                                                      |
| Resource Sizing        | 5     | Size by cluster/volume tier                                                                                      |
| Security               | 5     | Never store tokens in Git, use ESO                                                                               |
| GitOps/Lifecycle       | 9     | Pin versions, progressive rollouts, prune: false for DynaKube                                                    |
| Cluster Alerting       | 11    | Node health, OOMKill, CrashLoop, Dynatrace component health                                                      |
| Workload Monitoring    | 6     | CPU throttling, P99 SLOs, <code>arrayAvg()</code> for timeseries                                                 |
| Specialized Monitoring | 5     | NGINX via ConfigMap; Prometheus from inside the cluster, OTel Collector for new setups                           |
| Troubleshooting        | 9     | DynaKube status, connectivity, DQL-based diagnostics                                                             |

## References

- [DynaKube parameters \(DT docs\)](#)
- [DynaKube feature flags \(DT docs\)](#)
- [Set up Dynatrace on Kubernetes \(DT docs\)](#)
- [How Dynatrace Kubernetes monitoring works \(DT docs\)](#)
- [Quickstart for K8s setup \(DT docs\)](#)
- [Full observability deployment \(DT docs\)](#)
- [Kubernetes log monitoring \(DT docs\)](#)

- [Metadata enrichment \(DT docs\)](#)
- [Kubernetes app \(DT docs\)](#)
- [Operator + DynaKube troubleshooting \(DT docs\)](#)
- [Dynatrace Operator Helm chart \(Dynatrace GitHub\)](#)
- [Dynatrace Operator releases \(Dynatrace GitHub\)](#)
- [Operator 1.11.0 release notes \(DT docs\)](#)
- [Kubernetes tag setup \(DT docs\)](#)

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